

Ion-ion Drag in Weakly Coupled Yukawa (Screened-Coulomb)
Beam Plasma:
A Finite-Separation Binary-Collision Model and Molecular
Dynamics Tests

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Abstract

We have developed a semianalytic binary-collision model for drag on dilute streaming ions in a classical plasma interacting through a Yukawa potential. Collisions begin and end at a finite separation representing the local many-particle environment, rather than asymptotically infinite separation. We compare predicted drag curves of an Si-H system with classical molecular dynamics simulations performed using LAMMPS.

I. INTRODUCTION

Understanding drag force in plasmas is important for astrophysical problems such as accretion flows and protoplanetary disks and inertial confinement fusion [1][2][3]. To first order, the drag force can be approximated by the product of the collision cross section σ , particle flux rate Φ , and momentum per particle mv .

$$F = \Phi mv \sigma \quad (1)$$

Below the thermal velocity, the drag force is expected to be linear in velocity because the collision cross section and particle flux rate are approximately constant and related to the thermal velocity [citation needed].

$$v \ll v_{th}, \Phi \sim nv_{th}, \sigma \propto v_{th}^{-4} \quad (2)$$

$$F \propto v \quad (3)$$

Above the thermal velocity, the coulomb collision cross section scales as

$$\frac{Q_1 Q_2}{4\pi\epsilon_0 r_c} = \frac{1}{2}mv^2 \quad (4)$$

$$\sigma \propto r_c^2 \propto v^{-4} \quad (5)$$

In this regime the drag force for coulomb interactions follows

$$v \gg v_{th}, F \propto v^{-2} \quad (6)$$

Around the thermal velocity, the maximum in drag force is achieved, known as the Bragg peak, and has physical consequences. For a constant force (for example, due to radiation

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pressure), if an ion exceeds the Bragg peak, then it will continue to accelerate as a runaway ion [4].

Binary collision transport calculations based on momentum-transfer cross sections for screened interactions have previously been developed for weakly and moderately coupled plasmas [5] [6]. Classical molecular-dynamics calculations have also quantified the modification and displacement of the stopping maximum with increasing coupling [7] [8]. In this work, each binary encounter begins and ends at a finite separation, rather than at asymptotically infinite separation, and the resulting drag is tested directly against molecular dynamics simulations.

In this paper, we present a theory for the drag force on an ion species streaming through a weakly coupled plasma background. We use an improved formulation of the screening length. Section II derives our theory. Section III presents numerical tests to determine the convergence properties of the numerical integrals. Section IV presents our theory fit to molecular dynamics simulations.

II. OUR THEORY

In this work, we consider the problem of test silicon ions moving through a hydrogen background, which is relevant for the study of atmospheres of white dwarf stars.

$$n_{Si} \ll n_H \tag{7}$$

This theoretical treatment is short of the complete boltzmann equation with collision operator, but is constructed from first principles. We make the assumption that the plasma are perfectly collimated, where particles move parallel or antiparallel to bulk-drift axis, while speeds are thermally distributed.

A. Two Body Collision at Finite Initial Separation

We assume silicon is in thermal equilibrium with hydrogen, though this is not a physical requirement. Let the relative bulk velocity between silicon and hydrogen be v_b . Consider binary collision between silicon and hydrogen in the center of mass frame. The two body collision model is only accurate up to the scale of the typical interatomic separation,

55 hence we will not assume that the collision starts with the two bodies at infinite separation,
 56 but rather at finite separation The initial energy of this system is:

$$E_i = \frac{1}{2}m_H v_H^2 + \frac{1}{2}m_{Si} v_{Si}^2 + U(r_i) \quad (8)$$

57 In COM frame:

$$m_H v_H = -m_{Si} v_{Si} \quad (9)$$

58 The relative velocity is given by:

$$v = v_H - v_{Si} \quad (10)$$

59

$$v = -v_{Si} \left(\frac{m_{Si}}{m_H} + 1 \right) \quad (11)$$

60

$$v_{Si} = -\frac{m_H}{m_H + m_{Si}} v \quad (12)$$

61

$$v_H = -\frac{m_{Si}}{m_H + m_{Si}} v \quad (13)$$

62 Thus the energy simplifies to

$$E_i = \frac{1}{2}\mu v^2 + U(r_i) \quad (14)$$

Where $\mu = \frac{m_H m_{Si}}{m_H + m_{Si}}$

We wish to obtain the scattering angle to get the momentum transfer per collision. From conservation of angular momentum:

$$L = m_H r_H^2 \dot{\phi}_H + m_{Si} r_{Si}^2 \dot{\phi}_{Si}$$

63 In COM frame,

$$\dot{\phi}_H = \dot{\phi}_{Si} = \dot{\phi} \quad (15)$$

64

$$m_H r_H = m_{Si} r_{Si} \quad (16)$$

65 Define

$$r = r_H + r_{Si} \quad (17)$$

66 Therefore:

$$r_H = \frac{\mu}{m_H} r \quad (18)$$

67

$$r_{Si} = \frac{\mu}{m_{Si}} r \quad (19)$$

68

$$L = \left(\frac{1}{m_H} + \frac{1}{m_{Si}} \right) \mu^2 r^2 \dot{\phi} \quad (20)$$

$$L = \mu r^2 \dot{\phi} \quad (21)$$

The energy is given by

$$E = \frac{1}{2} \mu \dot{r}^2 + \frac{L^2}{2\mu r^2} + U(r) \quad (22)$$

We obtain

$$\dot{r} = \sqrt{\frac{2}{\mu} (E - U(r) - \frac{L^2}{2\mu r^2})} \quad (23)$$

And

$$\frac{d\phi}{dr} = \frac{\dot{\phi}}{\dot{r}} = \frac{L/(\mu r^2)}{\sqrt{\frac{2}{\mu} (E - U(r) - \frac{L^2}{2\mu r^2})}} \quad (24)$$

We introduce the impact parameter b as the least separation between Si and H if no force acts. b_H is the least separation between H and center of mass. For the rest of the derivation, consider the problem from perspective of H moving in COM frame.

$$b_H = b \frac{m_H}{m_H + m_{Si}} \quad (25)$$

$$\dot{\phi} = \frac{v_{H,\phi}}{r_H} \quad (26)$$

$$\frac{v_{H,\phi}}{v_H} = \frac{b_H}{r_H} = \frac{b}{r} \quad (27)$$

$$L = \mu r \frac{bv_H}{r_H} = \mu bv \quad (28)$$

We obtain

$$\frac{d\phi}{dr} = \frac{bv_i}{r^2 \sqrt{\frac{2}{\mu} (E - U(r) - \frac{\mu b^2 v_i^2}{2r^2})}} \quad (29)$$

Where b, v_i , and E are constants set by the initial conditions of the two body collision. For a general potential $U(r)$, from conservation of angular momentum and energy one obtains. This agrees with form obtained in [9].

$$\Delta\phi = 2 \int_{r_{max}}^{r_{min}} dr \frac{bv_i}{r^2 \sqrt{\frac{2}{\mu} (E - U(r) - \frac{\mu b^2 v_i^2}{2r^2})}} \quad (30)$$

Where $\Delta\phi$ is the angle between P_i and P_f (See FIG.1)

The scattering angle θ (deflection of initial velocity vector) is thus

$$\theta = \pi - \Delta\phi - 2\alpha \quad (31)$$

$$\sin \alpha = \frac{b_H}{r_H} = \frac{b}{r} \quad (32)$$

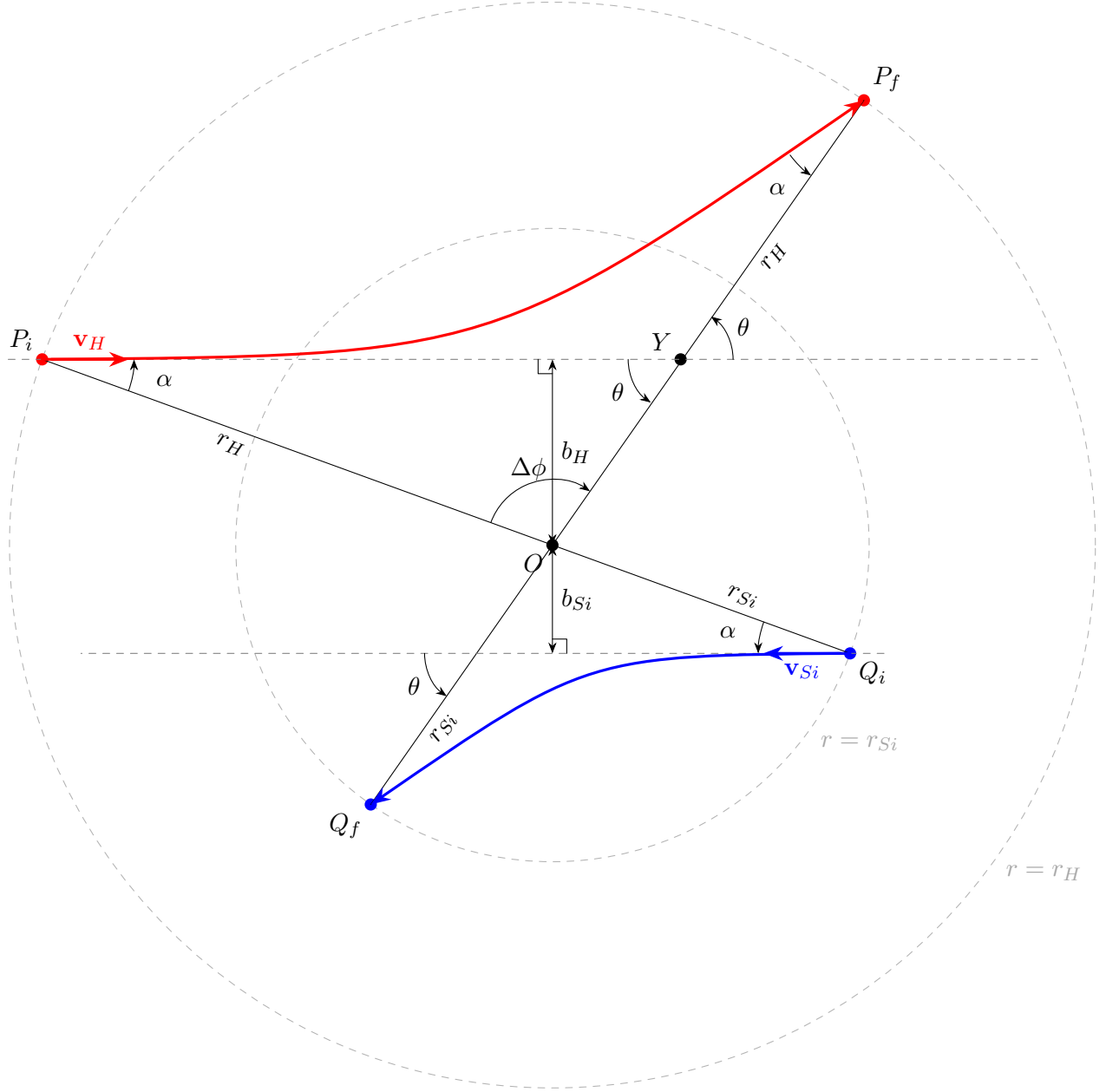


FIG. 1. Schematic of two body scattering with finite starting radius. Note that the trajectories are symmetric about a line connecting the point of closest approach and the center of mass. Using this symmetry, the initial and final velocity and position vectors are offset by angle α .

86 The distance of closest approach is characterized by a divergence of the integrand, and is
 87 given by

$$E - U(r_{min}) - \frac{\mu b^2 v_i^2}{2r_{min}^2} = 0 \quad (33)$$

B. Yukawa Potential

For the Yukawa potential:

$$U(r) = A \frac{\exp(-k_0 r)}{r}, A = \frac{Z_H^{eff} Z_{Si}^{eff} q_e^2}{4\pi\epsilon_0} \quad (34)$$

Where effective charges are due to ionization of atoms in the extreme plasma environment. The standard Saha equation is unreliable for high densities [citation needed]. We use state of the art DOE code to calculate the effective charges. First, the electron pressure and Z_H^{eff} in a hydrogen plasma with specified number density and temperature was calculated. Then, the Z_{Si}^{eff} was iterated to match the electron pressure of the previous calculation. See Table I for environmental conditions used

$$\Delta\phi_Y = bv_i \sqrt{2\mu} \int_{r_{min}}^{r_{max}} dr \frac{1}{r^2 \sqrt{E - A \exp(-k_0 r)/r - \frac{\mu b^2 v_i^2}{2r^2}}} \quad (35)$$

The denominator has a root at r_{min}

$$E - A \exp(-k_0 r_{min})/r_{min} - \frac{\mu b^2 v_i^2}{2r_{min}^2} = 0 \quad (36)$$

Let f be the integrand of equation 29.

$$\lim_{r \rightarrow r_{min}^+} f(r) \sim O((r - r_{min})^{-1/2}) \quad (37)$$

TABLE I. Effective charge calculated from DOE code. Effective charge of hydrogen calculated from density and temperature. Effective charge of silicon calculated based on density which matches the temperature and electron pressure values of Hydrogen. ρ_{Si} is not physically relevant; it is the equivalent density of Silicon needed to stand in for the effect of hydrogen to produce the same temperature and electron pressure conditions of background hydrogen; this is due to average ionization code only working for single species.

$\rho_H[g/cm^3]$	$\rho_{Si}[g/cm^3]$	$T[K]$	Z_H^{eff}	Z_{Si}^{eff}	Γ_{H-H}	Γ_{Si-Si}	$P[Mbar]$
1.0e-5	1.79e-4	5000	0.16	0.26	0.03	0.06	6.6e-7
1.0	7.45	5000	0.65	3.82	1.94	42.7	1.04
1.0e-5	6.2e-5	100,000	0.93	4.27	0.42	0.54	7.7e-5
1.0	7.20	100,000	0.68	3.81	1.05	21.0	2.48

100 This can be verified by checking that the derivative of the argument of the square root is
 101 nonzero at r_{min} :

$$A \exp(-k_0 r_{min}) \left(\frac{k_0}{r_{min}} + \frac{1}{r_{min}^2} \right) + \frac{\mu b^2 v_i^2}{r_{min}^3} > 0 \quad (38)$$

102 We make the further substitution:

$$r - r_{min} = x^2, dr = 2x dx \quad (39)$$

103 This gets rid of the divergence in the integrand:

$$f(r) dr = f(x^2 + r_{min}) (2x dx) \quad (40)$$

104

$$\lim_{x \rightarrow 0} f(x^2 + r_{min}) (2x dx) \sim O\left(\frac{x}{x} dx\right) \quad (41)$$

105 The complete integral becomes:

$$\Delta\phi_Y = 2bv_i \sqrt{2\mu} \int_0^{\sqrt{r_{max}-r_{min}}} \frac{x dx}{(r_{min} + x^2)^2 \sqrt{E - A \exp(-k_0(r_{min} + x^2)) / (r_{min} + x^2) - \frac{\mu b^2 v_i^2}{2(r_{min} + x^2)^2}}} \quad (42)$$

106 Using improved screening length from [10]

$$\lambda_S = \lambda_D \left(1 + \frac{2T_f}{3T} \right)^{1/4} \quad (43)$$

107 Where the electron Debye length is given by

$$\frac{1}{\lambda_D^2} = \sum_s \frac{n_s q_s^2}{\epsilon_0 k_B T_s} \quad (44)$$

108 So that

$$k_0 = \frac{1}{\lambda_S} \quad (45)$$

109 To close this section, we note that the divergence in the integrand of $\Delta\phi$ at r_{min} in a
 110 Yukawa potential is very different than in a Coulomb potential ($1/r$). It is tempting but
 111 incorrect to get rid of the divergence by using an analytically solvable Coulomb potential
 112 and numerically integrating the difference. One can check using L'Hopital's rule that when
 113 the strength of the Coulomb potential is tuned to match r_{min} with a Yukawa potential, the
 114 integrands diverge to infinity at different rates.

C. Drag Force

The momentum transfer per collision in the v_b direction experienced by the Silicon is thus

$$\Delta p = -m_{Si}v_{Si}(1 - \cos \theta) \quad (46)$$

$$\Delta p = -\mu v(1 - \cos \theta) \quad (47)$$

The drag force is the momentum transfer per collision multiplied by collision frequency, integrated over the velocity distribution and cross sectional area.

$$F_{Si} = \iint f(v)\omega_c(v, b)\Delta p(v, b)dv dA \quad (48)$$

$$F_{Si} = \iint f(v)\omega_c(v_{rel})\Delta p(v_{rel}, b)dv 2\pi b db \quad (49)$$

With collision frequency per unit area

$$\omega_c(v) = n_H v \quad (50)$$

It is possible to define an effective cross section a posteriori.

The 3D relative velocity distribution function is

$$f(\mathbf{v}) = \left(\frac{\mu}{2\pi k_B T}\right)^{3/2} \exp\left(-\frac{\mu}{2k_B T}((v_x - v_b)^2 + v_y^2 + v_z^2)\right) \quad (51)$$

Transverse relative velocities are ignored. We only retain the thermal velocity variations parallel to the bulk drift v_b .

$$f(v) = \left(\frac{\mu}{2\pi k_B T}\right)^{1/2} \exp\left(-\frac{\mu}{2k_B T}((v - v_b)^2)\right) \quad (52)$$

Therefore giving

$$F_{Si} = -2\pi n_H \mu \left(\frac{\mu}{2\pi k_B T}\right)^{1/2} \int_{-\infty}^{\infty} dv \int_0^{b_{max}} db \exp\left(-\frac{\mu(v - v_b)^2}{2k_B T}\right) |v| b v (1 - \cos(\theta(v, \rho))) \quad (53)$$

Note that we have neglected to consider angular variation in incident hydrogen velocity relative to bulk velocity (modelled velocity of incident particle as always in same direction and only considered hydrogen to come from forward plane). b_{max} and r_{max} correspond to choosing an upper limit for the interaction range. The justification is that in weakly coupled plasmas, the binary interactions do not extend to infinite range due to the presence of other background ions affecting interaction ranges above the typical interatomic spacing.

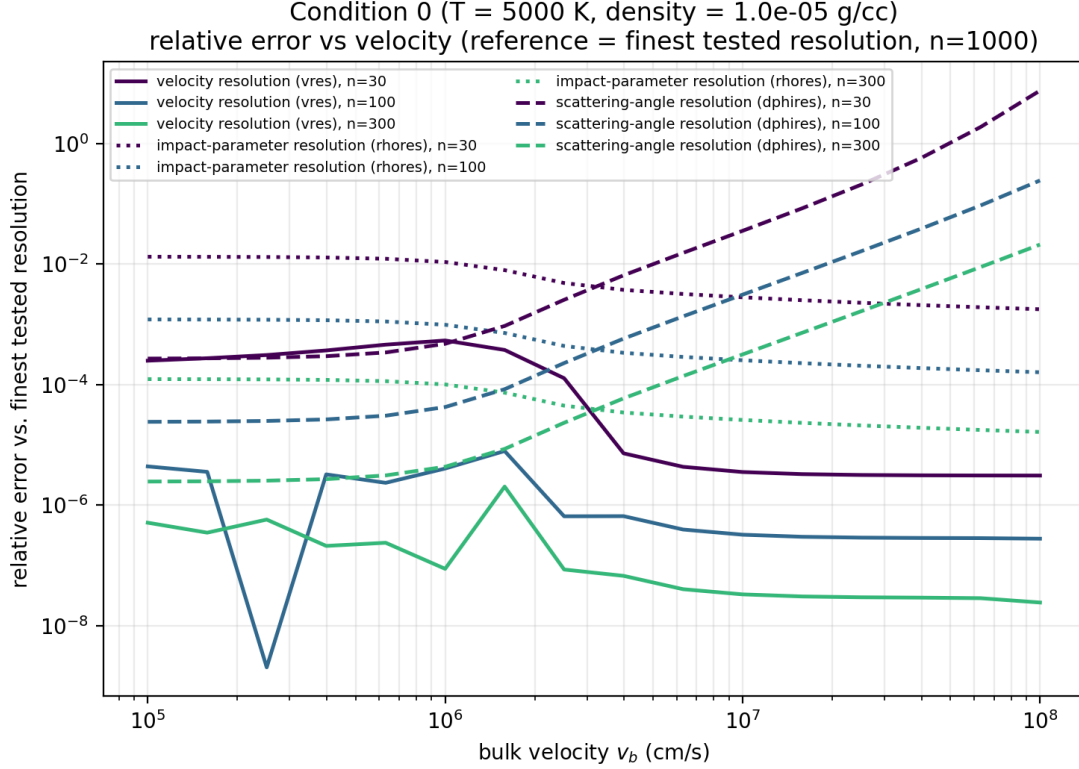


FIG. 2. Convergence test of numerical integration of plasma drag theory. Solid, dashed, and dotted lines correspond to resolution sweeps of relative velocity v , impact parameter b , and radius r , which is the variable integrated to obtain scattering angle $\Delta\phi$. Resolutions of 30, 100, 300, and 1000 were tested, with errors relative to the finest resolution of 1000 plotted as a function of velocity.

133 III. NUMERICAL STUDIES

134 Convergence tests were done on the numerical integration of the drag force, shown in
 135 Figure 2. Midpoint Euler integration was used. The scattering angle resolution refers to
 136 the radius r which is integrated to obtain $\Delta\phi$. These results motivated the choice of 30, 100,
 137 and 1000 as the default values for velocity, impact parameter, and scattering angle integral
 138 resolutions respectively. Setting the impact parameter upper bound equal to the interatomic
 139 spacing a_H :
 140

$$\frac{4}{3}\pi a_H^3 = \frac{1}{n_H} \quad (54)$$

$$b_{max} = a_H = \left(\frac{3}{4\pi n_H}\right)^{1/3} \quad (55)$$

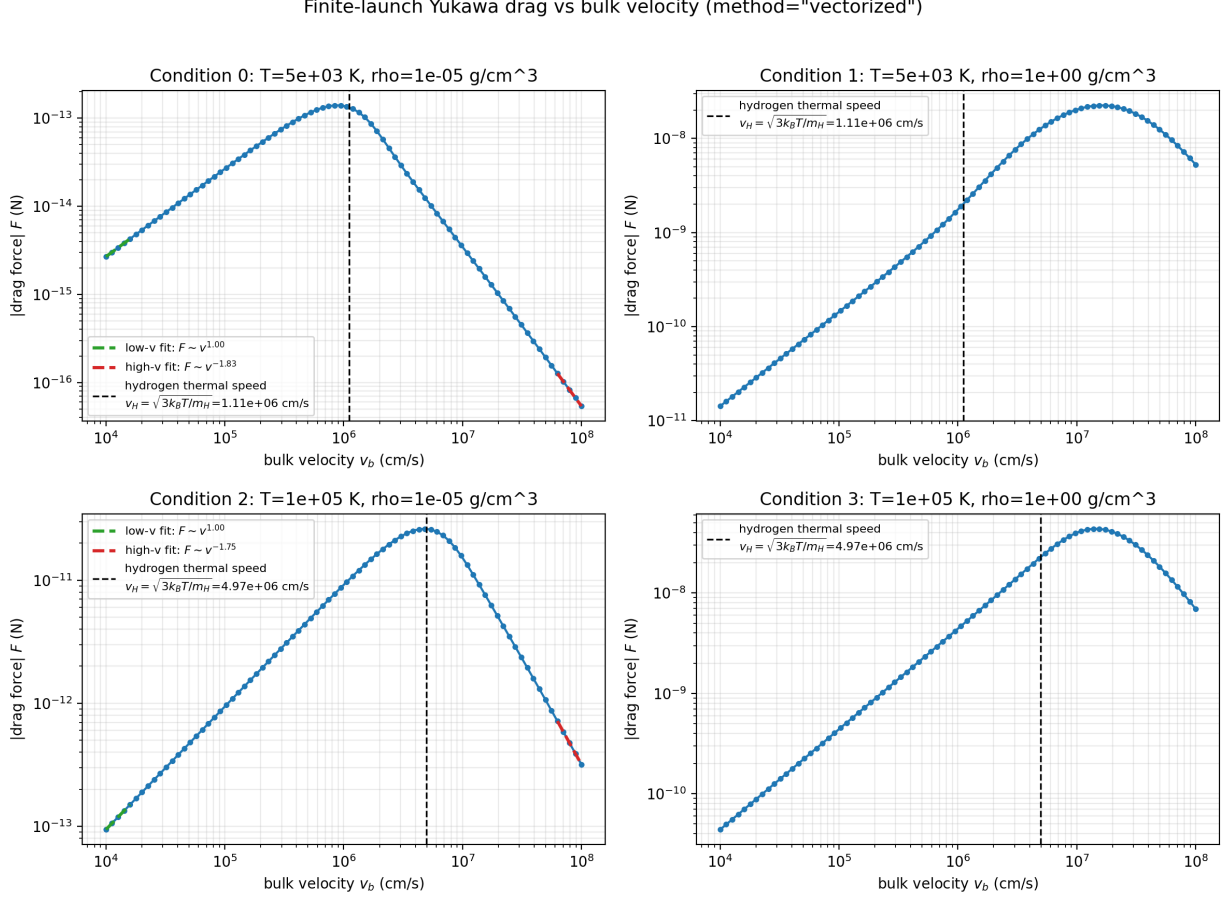


FIG. 3. Drag curves of this theory. All cases show linear drag in regime where bulk velocity much less than hydrogen thermal velocity (shown with black vertical line). Conditions 0 and 2 are the weakly coupled cases, whereas conditions 1 and 3 are the strongly coupled cases.

the drag curves in Figure 3 were obtained. For the weakly coupled cases (0,2), the Bragg peak is centered on the hydrogen thermal velocity, but the velocity of peak drag increases to above the thermal velocity. As expected, due to the use of the Yukawa rather than Coulomb potential, the high velocity regime does not have a v^{-2} power law.

IV. MOLECULAR DYNAMICS SIMULATIONS

A. Setup

Molecular dynamics simulations were run using LAMMPS [citation needed] to validate our theory. The timestep was chosen such that at the thermal or bulk velocity, a Hydrogen

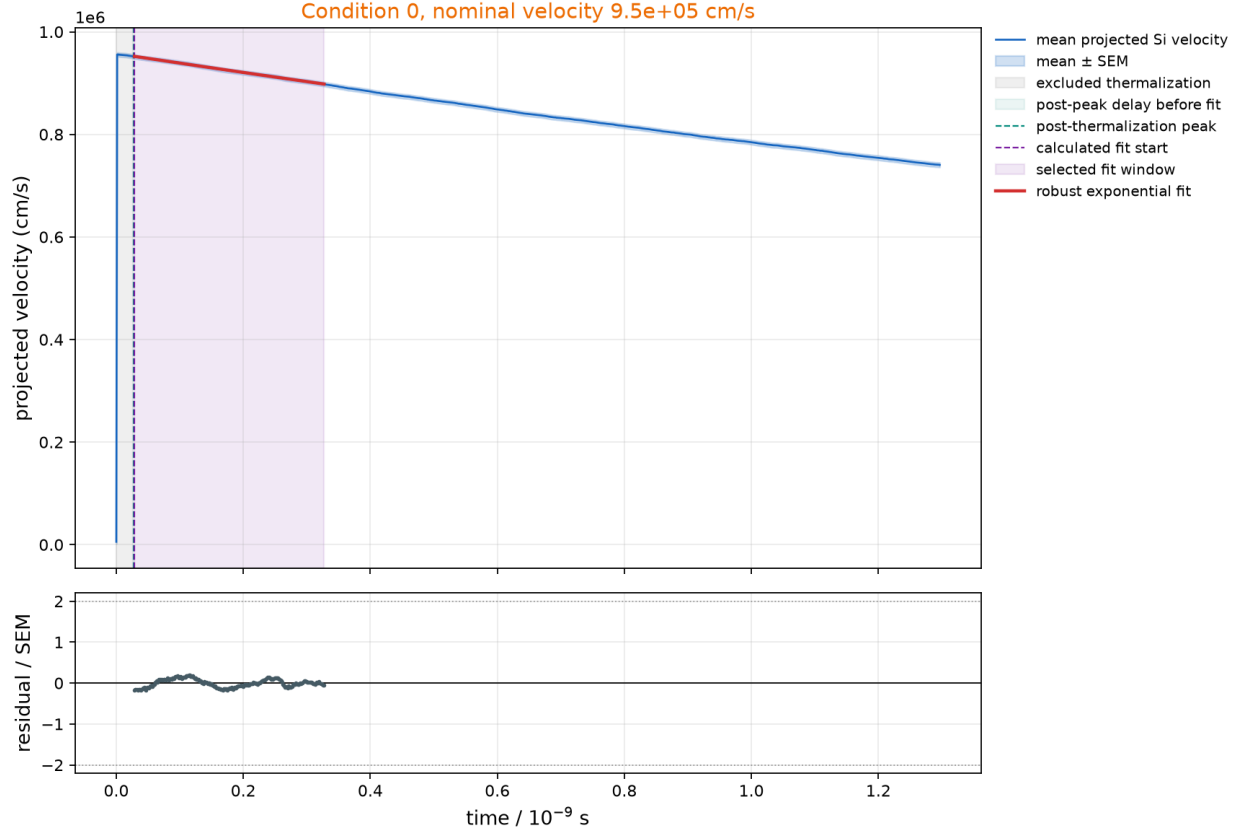


FIG. 4. Representative sample of LAMMPS MD data (blue line) and best fit (red).

atom travels at most 1% of the interatomic separation or screening length. NVE integrations were used. An alternative to NVE integration was tested, where streaming Si atoms were held at constant velocity and the acceleration that would have been applied to Si recorded. However, this approach resulted in very high errors and was physically dubious.

Each simulation involved 100,000 hydrogen and 1000 silicon atoms. A tradeoff exists between getting good statistics and maintaining uniform background conditions. The former requires increasing the Si-H ratio while the latter requires decreasing it. Due to using periodic boundary conditions, it was necessary to offset the initial velocities of the Si atoms at 0.3 radians to the z axis, to avoid particles following up on their own evacuated paths. To improve simulation speed, interactions with separation greater than cutoff radius of twice the screening length were ignored. Exponential functions were fit to the drag force as a function of velocity, as shown in Figure 4.

b_max fit vs LAMMPS ($r_l = a_H$ fixed)

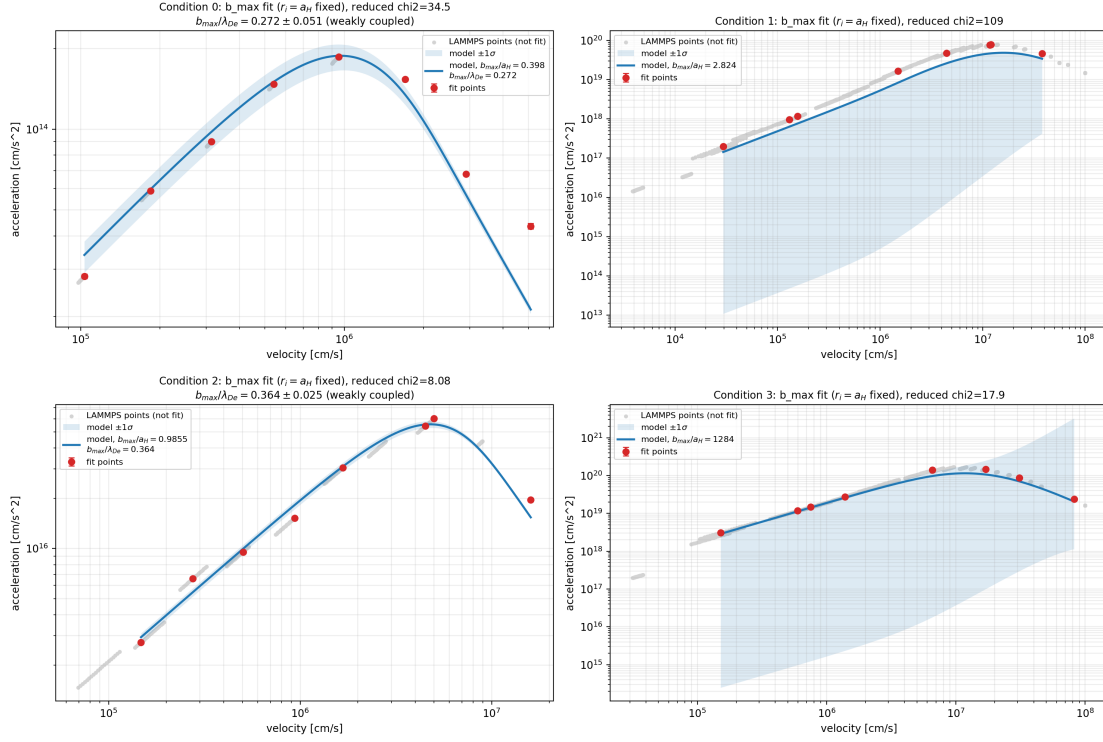


FIG. 5. Figure showing fit of best fit of $b_{\max}$ parameter in our theory for weakly coupled plasmas (left column) and strongly coupled plasmas (right column).

B. Comparison of Theory and Molecular Dynamics Simulations

Figure 5 shows the best fit of the parameter $b_{\max}$ for two weakly coupled and two strongly coupled cases. For the weakly coupled cases, the debye length seems to be a more consistent reference for $b_{\max}$ than the interatomic spacing. For the strongly coupled cases, $b_{\max}$ is much greater than the interatomic spacing, corresponding to the significance of long range interactions due to poor screening. The strongly coupled cases were included to test the model's range of applicability and is not the intended application of this theory.

V. CONCLUSION

In this work, we have presented a numerical theory derived from first principles with a single fitted parameter, $b_{\max}$, which fits MD simulations well around and below the Bragg peak.

175 Future work will involve validating this theory against experiments. Another aspect of
 176 improvement is to integrate over all three velocity dimensions, and evaluate whether an
 177 accuracy improvement is achieved that justifies the additional computational cost.

178 **Appendix A: Literature Review**

What we call drag is referred to in literature by terms such as "dynamical friction",
 "friction", or "stopping power" Suprathermal ions mostly slowed down by collisions with
 electrons.

$$\Lambda = \frac{\lambda_D}{b_{\frac{\pi}{2}}}$$

Where Debye length λ_D is defined (dropping ion contribution):

$$\lambda_D = \sqrt{\frac{\epsilon_0 k_B T_e}{n_e q_e^2}}$$

Depending on notation of author, impact parameter of 90° scattering is

$$b_0 = b_{\frac{\pi}{2}} = \frac{qQ}{4\pi\epsilon\mu u^2}$$

179 Where u is relative velocity and $\mu = \frac{mm'}{m+m'}$ is reduced mass. We proceed with a brief review
 180 of the existing literature on dynamical friction force in plasmas. The scope is limited to
 181 complete theories that treat the Bragg peak region, not just the extreme velocity limits.

182 **1. Gurnett and Bhattacharjee**

Drag force on test species s due to s' [11]

$$\mathbf{F}_s \mathbf{s}' = -\Gamma_{ss}. n_0 [erf(y) - y \frac{d}{dy} erf(y)] \frac{\mathbf{v}}{v^3}$$

Where

$$\Gamma_{ss} = \frac{e_s^2 e_s'^2}{4\pi\epsilon_0^2 m_s \mu_{ss'}} \ln \Omega_s$$

183 **2. Woods**

Gives acceleration [12]

$$\begin{aligned} < \Delta w > = -\frac{n_i \Gamma}{w^2} \hat{\mathbf{w}} \\ \Gamma &= \frac{(ZZ'e^2)^2 \ln \Lambda}{4\pi\epsilon_0^2 m^2} \end{aligned}$$

3. Frank, King, Raine

Give slowing down timescale [13]

$$t_s = -\frac{U}{\frac{dU}{dt}}$$

$$t_s \approx \frac{[4\pi\epsilon_0]^2 m_1^2 U^3}{8\pi N e_1^2 e_2^2 \ln \Lambda} \frac{m_2}{m_1 + m_2}$$

4. Bellan

[14]

$$\frac{\partial u}{\partial t} = \frac{n_e e^2 \ln \Lambda}{4\pi\epsilon_0^2} \frac{q_T^2}{m_T^2} \left[Z \left(1 + \frac{m_T}{m_i} \right) \frac{m_i}{2k_B T_i} \frac{d}{d\epsilon_i} \left(\frac{\text{erf}(\epsilon_i)}{\epsilon_i} \right) \right. \\ \left. + \left(1 + \frac{m_T}{m_e} \right) \frac{m_2}{2k_B T_e} \frac{d}{d\epsilon_e} \left(\frac{\text{erf}(\epsilon_e)}{\epsilon_e} \right) \right]$$

Where

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x \exp(-w^2) dw$$

$$\epsilon_{i,e} = \sqrt{\frac{m_{i,e}}{2k_B T_{i,e}}} u$$

5. Pert

For isotropic test particle distribution $f(v)$ and background particle distribution $f'(\mathbf{v})$, Dynamical friction equal to momentum flux [15]

$$\mathbf{J} = -\frac{4\pi Z^2 Z'^2 e^4 n'}{(4\pi\epsilon_0)^2 m'} \ln \Lambda [\text{erf}(\zeta) - \zeta \text{erf}(\zeta)] \\ \times \left(f(v) + \frac{m' v_T'^2}{2mv} \frac{df(v)}{dv} \right) \frac{\mathbf{v}}{v^3}$$

Where

$$\zeta = \frac{v}{v_T'}, v_T' = \sqrt{\frac{2k_B T}{m'}}$$

6. Dreicer

For force on electrons due to electron-ion collisions, in limit of background ion gas at zero temperature [4]:

$$-eE_c \Psi(z) = -eE_c \frac{\epsilon_2(z) - z \frac{d\epsilon_2}{dz}}{z^2}$$

Where

$$z = \beta_e^{\frac{1}{2}} |v_e - v_i|$$

$$\beta_e = \frac{m_e}{2kT_e}$$

$$eE_c = nm\Gamma_e\beta_e$$

$$\epsilon_2(z) = \frac{2}{\sqrt{\pi}} \int_0^z \exp(-t^2) dt$$

$$\Gamma_e = 4\pi \left(\frac{e^2}{4\pi\epsilon_0 m_e} \right)^2 \log\left(\frac{\lambda}{p_0}\right)$$

188 Where λ is Debye radius, p_0 is impact parameter for 90 degree coulomb deflection

189 7. Sprenkle et al

[6]

$$F_{ss'} = -\frac{16}{3} \frac{\sqrt{\pi} e^4 n_{s'}}{(4\pi\epsilon_0)^2 m_{ss'} v_{ss'}^3} \Xi(\Delta\bar{V})(\mathbf{u}_s - \mathbf{u}_{s'})$$

Where

$$\Delta\bar{V} = \frac{|\mathbf{u}_s - \mathbf{u}_{s'}|}{v_{ss'}}$$

$$m_{ss'} = \frac{m_s m_{s'}}{m_s + m_{s'}}$$

$$v_{ss'} = \sqrt{\frac{2k_B T_s}{m_s} + \frac{2k_B T_{s'}}{m_{s'}}}$$

$$\Xi(\Delta\bar{V}) = \frac{3}{16} \frac{1}{\Delta\bar{V}^3} \frac{1}{2} \int_0^\infty d\xi x i^2 \frac{\sigma_{ss'}^{(1)}(\xi)}{\sigma_0} \chi$$

$$\chi = (2\xi\Delta\bar{V} + 1)e^{-(\xi+\Delta\bar{V})^2} + (2\xi\Delta\bar{V} - 1)e^{-(\xi-\Delta\bar{V})^2}$$

$$v_{T_s} = \sqrt{\frac{2k_B T_s}{m_s}}$$

$$\sigma_0 = \frac{\pi e^4}{(4\pi\epsilon_0)^2 m_{ss'}^2 v_{ss'}^4}$$

For coulomb cross section obtain

$$\Xi(\Delta\bar{V}) = \frac{3\sqrt{\pi}}{4} \frac{\phi(\Delta\bar{V}^2)}{\Delta\bar{V}^3} \ln \Lambda$$

$$\psi(x) = \text{erf}(\sqrt{x}) - \frac{2}{\sqrt{\pi}} \sqrt{x} e^{-x}$$

For Yukawa cross section [5]

$$\sigma_{ij}^{(n)}(w, \lambda) = 2\pi\lambda^2 \phi_n(w)$$

$$\phi_n(w) = \int_0^\infty d\rho \rho [1 - \cos^n[\theta_{ij}(\rho, w)]]$$

Fit asymptotic forms for strongly coupled $w > 1$

$$\phi^{SC}(w) = \frac{c_0 + c_1 \ln(w) + c_2 \ln^2(w) + c_3 \ln^3(w)}{1 + c_4 \ln(w)}$$

and weakly coupled $w < 1$ limits

$$\phi^{WC}(w) = \frac{n}{2w^4} \ln(1 + w^2) P(w)$$

$$P(w) = \frac{d_0 + d_1 \ln(w) + d_2 \ln^2(w) + \ln^3(w)}{d_3 + d_4 \ln(w) + d_5 \ln^2(w) + \ln^3(w)}$$

190 8. Li and Petrasso

Give stopping power accounting for plasma oscillations [2]

$$\begin{aligned} \frac{dE^{t/f}}{dx} \approx & -\frac{Z_t e^2}{v_t^2} \omega_{pf}^2 \{G(x^{t/f}) \ln \Lambda_b \\ & + \Theta(x^{t/f}) \ln[1.123(x^{t/f})^{1/2}]\} \end{aligned}$$

Where $\Theta(x^{t/f})$ is step function whose value is 0 (1) for $x^{t/f} \leq 1 (> 1)$,

$$x^{t/f} = \frac{v_t^2}{v_f^2}$$

$$G(x^{t/f}) = \mu(x^{t/f}) - \frac{m_f}{m_t} \left\{ \frac{d\mu(x^{t/f})}{dx^{t/f}} - \frac{1}{\ln \Lambda_b} \left[\mu(x^{t/f}) + \frac{d\mu(x^{t/f})}{dx^{t/f}} \right] \right\}$$

Field plasma frequency

$$\omega_{pf} = \sqrt{\frac{4\pi n_f e_f^2}{m_f}}$$

Maxwell Integral

$$\mu(x^{t/f}) = \frac{2}{\sqrt{\pi}} \int_0^{x^{t/f}} e^{-\xi} \sqrt{\xi} d\xi$$

$$\Lambda_b = \frac{\lambda_D}{p_{min}}$$

$$p_{min} = \sqrt{\left(\frac{e_t e_f}{m_r u^2}\right)^2 + \left(\frac{\hbar}{2m_r u}\right)^2}$$

191 where λ_D is the debye length and m_r is the reduced mass and u is the relative velocity

[7] Semianalytic model with fit parameters, building on foundation of [16]

$$\frac{d\bar{E}}{dx} \approx -R(w)[G(w) \ln(e^{1/2} + \frac{\alpha + w^2}{g_0}) + H(w)]$$

$$R(w) = \frac{[M_1 + bM_2(w)w^2](1+g)^{2/3}}{w^2(1+bw^2)}$$

$$M_1 = s \frac{\ln(1 + \alpha e^{-1/2}/[g(1 + aZ^2g)])}{\ln(1 + \alpha e^{-1/2}/g_0)}$$

$$M_2(w) = \frac{1}{s^2} \frac{\ln(1 + s^3 w^3/g)}{\ln(1 + w^3/g_0)}$$

$$G(w) = \text{erf}\left(\frac{w}{\sqrt{2}}\right) - \sqrt{\frac{2}{\pi}} w e^{-w^2/2}$$

$$H(w) = \frac{w^4 \ln w}{12 + w^4} - \frac{w^3}{3\sqrt{2}\pi} e^{-w^2/2}$$

Intratarget coupling parameter, target charge q_e

$$\Gamma = \frac{q_e^2}{r_0 T}$$

Wigner-Seitz radius

$$r_0 = \left(\frac{4\pi n_e}{3}\right)^{-1/3}$$

Thermal velocity

$$v_{th} = \sqrt{T/m_e}$$

$$g = \sqrt{3}|Z|\Gamma^{3/2}$$

$$\tilde{v} = \frac{v}{v_{th}(1+g)^{1/3}}$$

$$\frac{d\tilde{E}}{dx} = \frac{dE}{dx} \frac{(1+g)^{2/3}}{(Z^2 q_e^2 / \lambda_D^2)}$$

Where

$$\alpha = 4 \exp(-2\gamma)$$

γ is Euler's constant 0.577216..

$$s = d(1 + cg)^{1/3}$$

$$w = \frac{v}{v_{th}s}$$

¹⁹³ Where fit parameters are $a = 1.04102 \times 10^{-5}$, $b = 0.183260$, $c = 0.116053$, $d = 0.824982$,

¹⁹⁴ and $g_0 = 2.03301 \times 10^{-3}$

Effective potential theory [8]

$$\begin{aligned}\frac{d\epsilon_F}{dx}|_{EPT} &= -m_\alpha V_0 \nu_\alpha \\ \nu_\alpha &= \frac{16\sqrt{\pi} n q^4}{3m_\alpha \mu v_T^3} \Xi_\alpha^{[F]}(\bar{V}_0) \\ \bar{V}_0 &= \frac{V_0}{v_T}\end{aligned}$$

velocity-dependent generalized coulomb logarithm

$$\begin{aligned}\Xi_\alpha^{[F]}(\bar{V}_0) &= \frac{3}{32} \frac{1}{\bar{V}_0^3} \int_0^\infty d\xi \xi^2 \frac{\bar{\sigma}(\xi)}{\sigma_0} \\ &\times [(2\xi \bar{V}_0 + 1)e^{-(\xi + \bar{V}_0)^2} + (2\xi \bar{V}_0 - 1)e^{-(\xi - \bar{V}_0)^2}] \\ v_r &= \xi v_T \\ \sigma_o &= \frac{\pi q^4}{\mu^2 v_T^4} \\ \bar{\sigma}(\xi) &= 2\pi \int_0^\infty db b [1 - \cos(\pi - 2\Theta(b, \xi v_T))] \\ \Theta(b, v_r) &= b \int_{r_0}^\infty dr \frac{1}{r^2} \left[1 - \frac{b^2}{r^2} - \frac{2\phi(r)}{\mu v_r^2} \right]^{-1/2}\end{aligned}$$

Where V_0 is projectile speed

[17] Where

$$\begin{aligned}\phi(x) &= \frac{2}{\sqrt{\pi}} \int_0^x e^{-y^2} dy \\ \frac{\partial V}{\partial t} &= \Gamma \frac{\partial H(V)}{\partial V} = -\nu_f(V) V \\ \nu_f &= \frac{2\Gamma}{V} \sum_s z_s^2 \left(\frac{m + m_s}{m_s} \right) n_s a_s^2 \Psi(a_s V) \\ \Psi(x) &= \frac{\phi(x) - x\phi'(x)}{2x^2} \\ a_s^2 &= \frac{m_s}{2k_B T_s}\end{aligned}$$

z, z_s are atomic numbers of scattered and scattering particles respectively

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